Project: Thermostats and barostats

Goal: To write **subroutines/functions** able to keep constant temperature and pressure for a given MD code, by Berendsen's method

- Data files + codes (from practice 2):
  - A molecular dynamics Lennard-Jones code, modeling the microcanonical ensemble
  - Data files and dimensions
  - First configuration
- It should be remembered that the code works internally in reduced units

Questions:

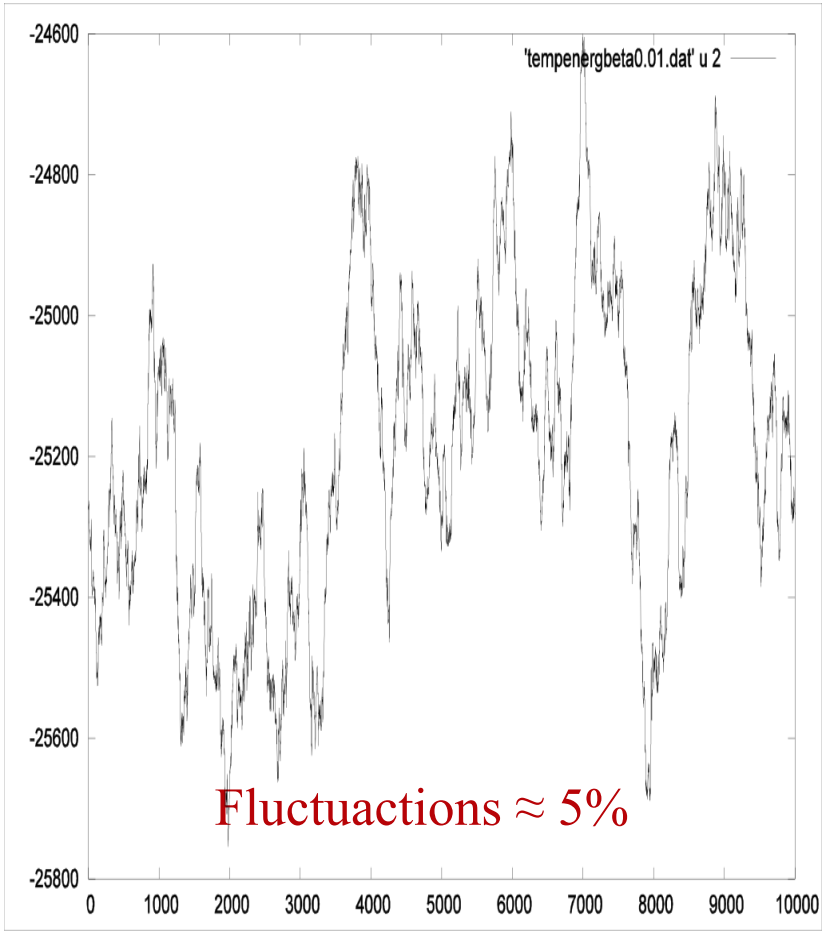
1.How do Berendsen's compare with Andersen's? (Accuracy, efficiency, speed of calculation,...). HINT: Modify the code and codify the Andersen thermostat.

2.Show that Berendsen's scaling is equivalent to a temperature coupling of the system with a heat bath at $T=T_0$, where $J = \alpha(T_0 - T)$. J is the heat flux and α is the heat transfer coefficient.

3. Derive the temperature (λ) and pressure (μ) scaling factors

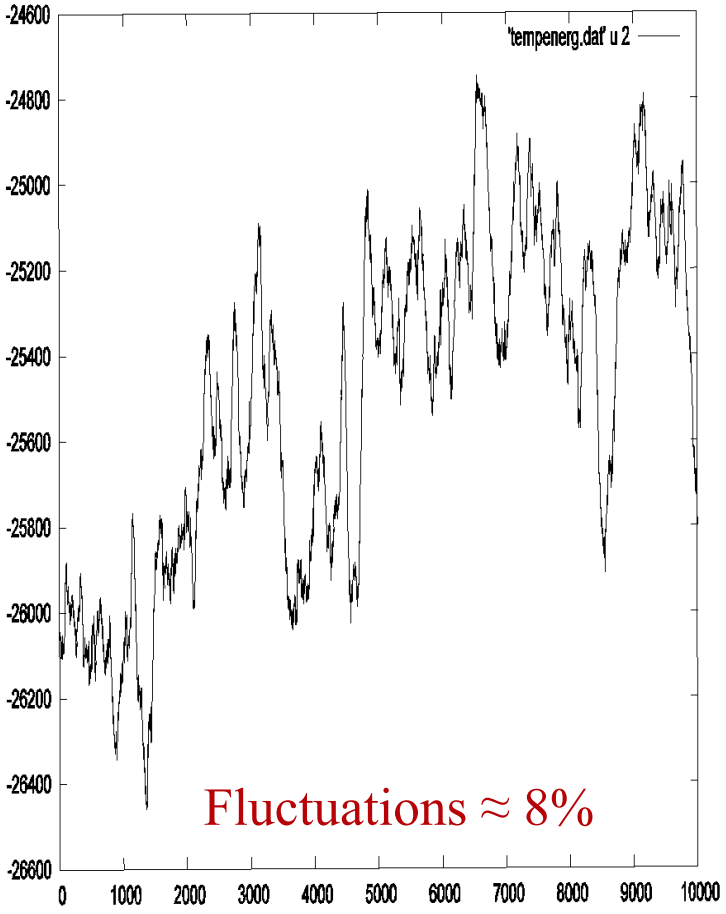
- **Results (Berendsen): Total energy**

NVE ensemble



Number of time steps

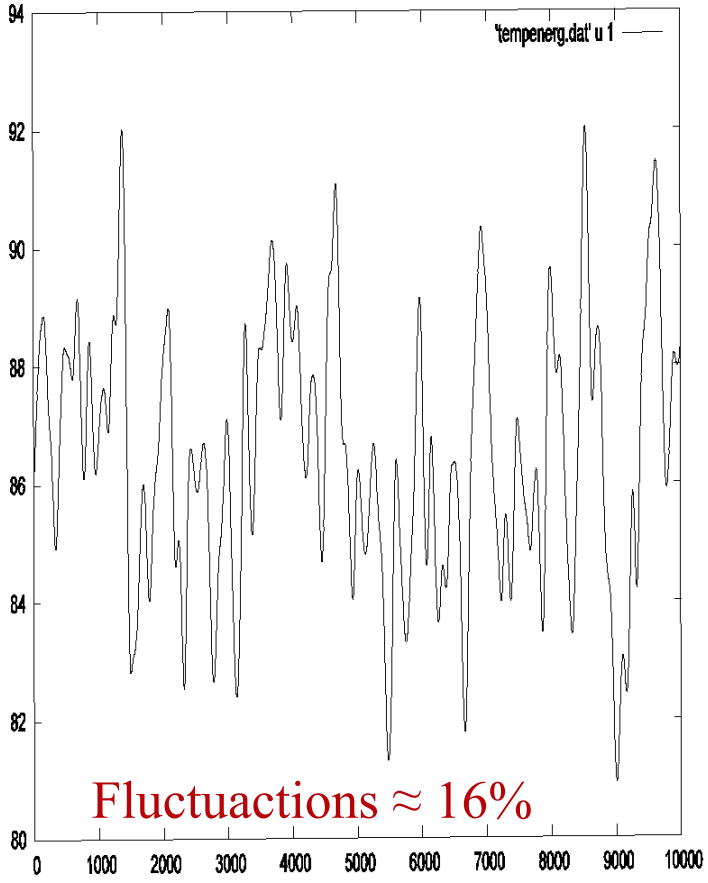
NPT ensemble



Number of time steps

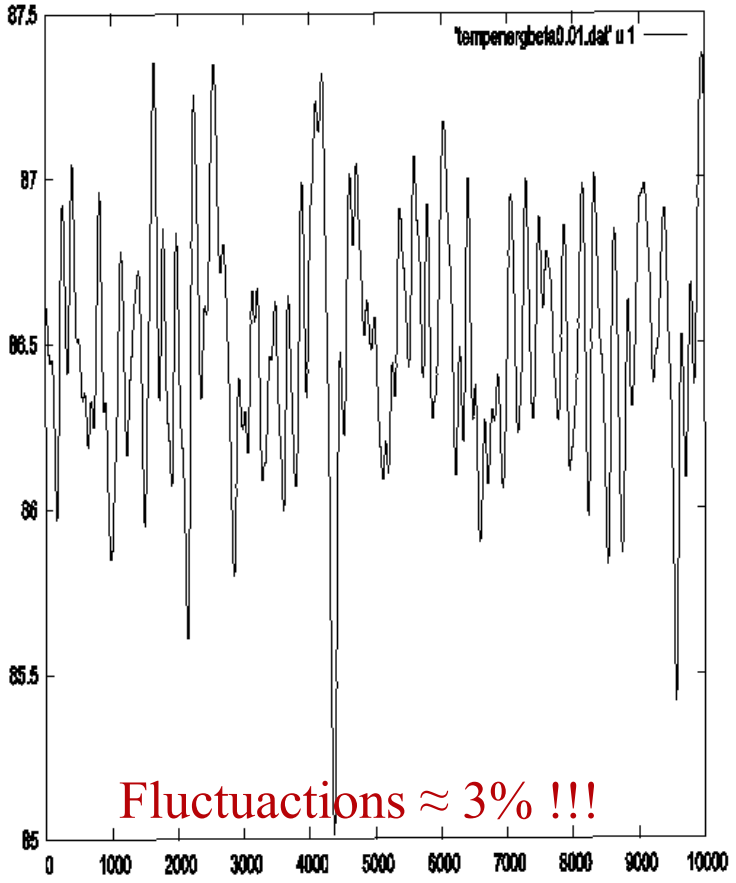
- **Results: temperature**

NVE ensemble



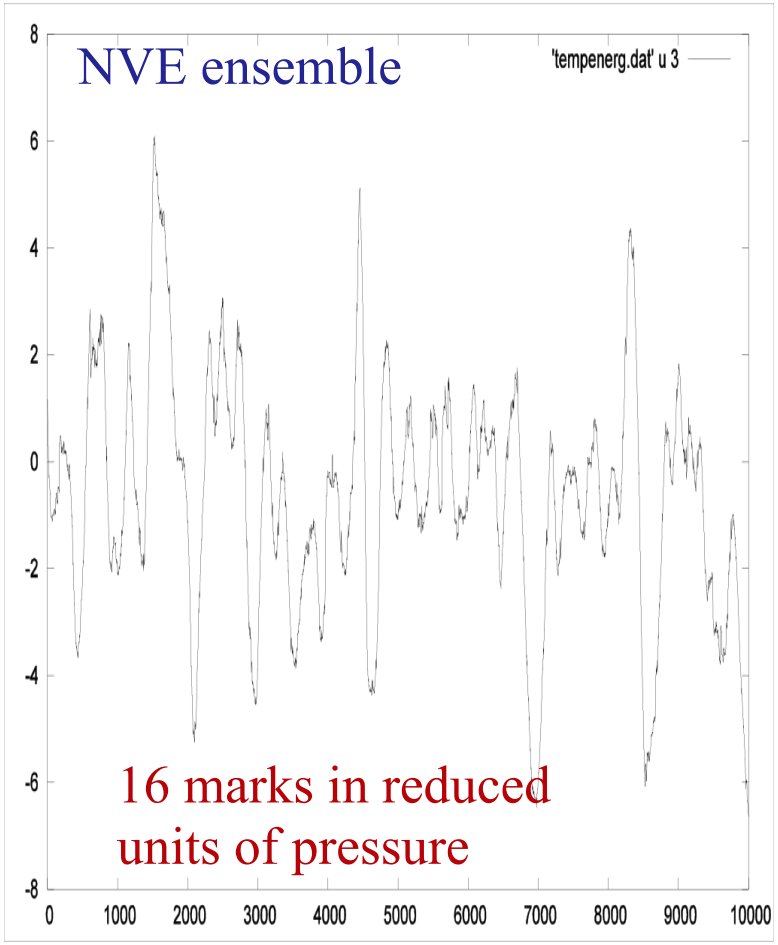
Number of time steps

NPT ensemble

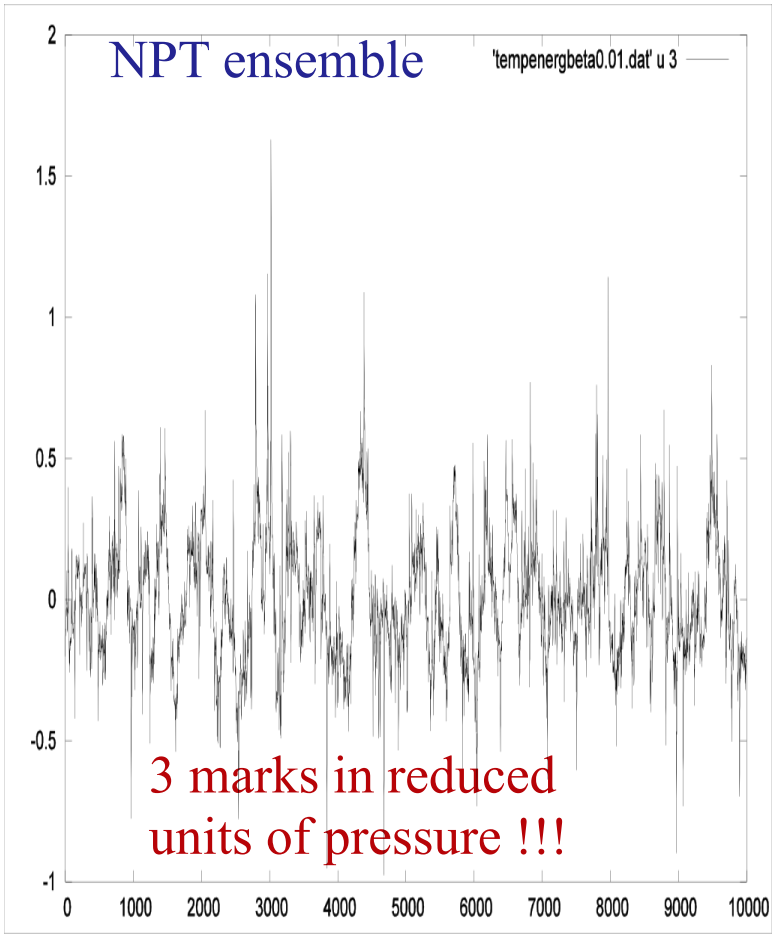


Number of time steps

- **Results: Pressure**



Number of time steps



Number of time steps